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Sequential Predictions of a Persistent Hidden Variable

A minimal completion of the Donadi–Hossenfelder outcome rule, and what repeated measurements of one qubit would show

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Summary

Donadi and Hossenfelder select the outcome of a single measurement by comparing a uniformly distributed hidden variable with the Born probability [1]. Deviations from the Born rule require the hidden variable to be sampled nonuniformly, for instance because it changes slowly between measurements, but the model specifies neither how it evolves nor an outcome rule free of the arbitrary ordering of basis states. This note supplies both in minimal form. Two ordering-free completions anchor the hidden variable either to the qubit or to the detector, and a renewal law with gap persistence $w = s_m e^{-\Delta/\tau}$ keeps every single-measurement marginal exactly at its Born value, so all predictions are serial correlations. Several results follow:

1. For detector-anchored persistence the most sensitive probe is two readouts of a twice-prepared balanced superposition: the outcome correlation equals w, first order in persistence. Qubit-anchored persistence turns with every pulse, so a re-preparation made of pulses removes that correlation. It survives in the orthogonal triad, whose endpoint correlation $s_m^2 e^{-\Delta/\tau}$ needs no reset, and in a first-order deviation of two readouts at small angles.
2. Randomizing which eigenstate each readout maps to the bright detector outcome leaves qubit-anchored persistence unchanged once the analysis tracks the frame that each relabeling pulse leaves behind, and removes detector-anchored persistence from the balanced tests, together with conventional memory carried by the detector or the motion.
3. Both completions couple outcomes through a shared threshold, which puts a cusp or kinks in the dependence on the preparation angle. When the angle enters only as a frame phase on a few levels, no quantum model with memory produces them (Proposition 2), so the shape can identify the hidden variable, at several times the persistence needed to detect it.
4. Detector-anchored persistence, the placement closest to Hossenfelder's description, matches conventional detector memory at fixed angles; the cusp separates the two. Raw records from photon-counting random-number generators bound it only down to the size of afterpulsing and dead-time effects, which produce the same fixed-angle correlation.

The qubit-anchored rule is the Bell–Mermin model. Using these rules to complete the Donadi–Hossenfelder model, and the persistence law, are constructions made for this note; they are not proposals of Donadi or Hossenfelder, and they do not address her 2025 collapse model. Draft v0.2 corrects the qubit-anchored predictions of v0.1, which treated re-preparation as leaving the hidden vector unrotated (Section 4, item 4).

1 Purpose and scope

The white paper adapts Hossenfelder's recurrence proposal [2, 3] into a same-qubit sequence $Z \to N_\theta \to Z$ and interprets data against an operational template, $Ae^{-\Delta/\tau} \sin^2\theta$, whose parameters await a specific dynamical theory (white paper Section 3.4). The co-authored toy model of Donadi and Hossenfelder comes closest to such a theory [1]. Its closing paragraph suggests that if the hidden variables are set by slowly changing detector degrees of freedom, repeated measurements of noncommuting observables could keep sampling the same region and

show correlations stronger than quantum mechanics predicts; the authors also call the model, strictly speaking, untestable, because it leaves the origin of the hidden-variable distribution open. Hance and Hossenfelder name a time correlation of measurement outcomes as the expected signature [4].

This note asks which minimal additions make that suggestion predictive for one qubit, and what the predictions imply for experimental design. Its scope is the persistence of an outcome-determining variable together with the outcome rule. For a single qubit, any model of this kind is a hidden-variable model with memory. The violation of statistical independence that makes the parent model superdeterministic appears only in Bell scenarios, which Section 8 leaves open. Hossenfelder's 2025 collapse model is outside the scope: its hidden variables are independent for each end state and its consecutive-rotation probabilities factorize [5].

2 The single-measurement rule

For two outcomes, Donadi and Hossenfelder draw one complex hidden variable λ uniformly from the unit disk and select outcome 2 if and only if $|\lambda|^2 > |\langle\chi_1|\psi\rangle|^2$, where $\{|\chi_1\rangle, |\chi_2\rangle\}$ is the detector eigenbasis and $|\psi\rangle$ the state [1]. Since $u \equiv |\lambda|^2$ is uniform on $[0, 1]$, each single measurement reproduces the Born rule. Two ingredients are missing for a sequence:

1. an update law for u between measurements;
2. an outcome rule independent of which basis state is listed first. With u held fixed through a Z, X, Z chain, every step compares u with $1/2$, so all three outcomes carry the same listing label, and the predicted middle correlation depends on that arbitrary listing.

3 Two ordering-free completions

Write the qubit state as a Bloch vector $\boldsymbol{s}$ and a measurement axis as a unit vector $\boldsymbol{n}$, so that quantum mechanics gives $\mathbb{P}(x = \pm 1) = \frac{1}{2}(1 \pm \boldsymbol{s} \cdot \boldsymbol{n})$ and the post-measurement state $x\boldsymbol{n}$. An atomic-qubit readout measures along $\boldsymbol{n}$ by applying a pulse R with $R\boldsymbol{n} = \hat{\boldsymbol{z}}$, detecting fluorescence, and labeling the result. The pulse fixes which eigenstate, $\sigma\boldsymbol{n}$ with $\sigma = \pm 1$, appears bright.

Detector-anchored completion (D). The hidden variable $u \in [0, 1]$ belongs to the detector, which compares it with the probability of its own bright outcome:

$$\text{bright} \iff u < p_b = \tfrac{1}{2}\,(1 + \sigma\,\boldsymbol{s} \cdot \boldsymbol{n})\,, \qquad x = \begin{cases} \sigma & \text{bright,} \\ -\sigma & \text{dark.} \end{cases} \tag{1}$$

The physical bright outcome fixes the listing, so the rule is well defined; its predictions depend on σ, which is a physical setting.

Qubit-anchored completion (Q). The hidden variable is a unit vector $\boldsymbol{\lambda}$, isotropically distributed, that transforms with the qubit under every pulse. The detector reads $\hat{\boldsymbol{z}}$ after R, so

$$x = \operatorname{sgn}\big[(R\boldsymbol{\lambda} + R\boldsymbol{s}) \cdot \hat{\boldsymbol{z}}\big] = \operatorname{sgn}\big[(\boldsymbol{\lambda} + \boldsymbol{s}) \cdot \boldsymbol{n}\big]. \tag{2}$$

Because $\boldsymbol{\lambda} \cdot \boldsymbol{n}$ is uniform on $[-1, 1]$, $\mathbb{P}(x = +1) = \frac{1}{2}(1 + \boldsymbol{s} \cdot \boldsymbol{n})$. Reversing $\boldsymbol{n}$ reverses x, so no listing enters. For each fixed axis, $u_{\boldsymbol{n}} = \frac{1}{2}(1 - \boldsymbol{\lambda} \cdot \boldsymbol{n})$ is uniform and Eq. (2) coincides with the original rule; the axes are tied together through one vector. Equation (2) is the Bell–Mermin model [6, Eq. 9], so named in Ref. [7]. Two relatives differ in ways that matter here. Bell's original construction uses one scalar for every axis [8]; it has the same single-measurement statistics but different sequential predictions under persistence. The Kochen–Specker model is the Bell–Mermin model with the redundant variable removed and redraws its hidden point after every measurement, so it has no persistence [7, 9]. The extension here needs the vector form, in which the hidden variable is independent of the state.

Both completions are deterministic given the hidden variable and reproduce the Born rule for any single measurement on a fresh draw. They differ in where the hidden variable lives. Completion D places it in the detector, the reading closest to Hossenfelder's description of hidden variables as degrees of freedom of the detector [2, 3]. Completion Q places it with the qubit, closer to the earlier question of hidden order in the atom itself [10].

4 Persistence

Between consecutive measurements, the hidden variable is kept with probability w_k and otherwise redrawn from its uniform distribution, independently of everything else. With redraws at rate $1/\tau$ during idle time and survival probability s_m per measurement,

$$w_k = s_m\, e^{-\Delta_k/\tau}, \tag{3}$$

which is the white paper's intervention-resolved template $A_{\text{eff}} = A_0 s_m^{n_m} e^{-\Delta/\tau_{\text{idle}}}$ with $A_0 = 1$. Several properties follow directly:

1. The uniform distribution is stationary, so every single-measurement marginal equals its Born value at every delay. Only joint statistics of two or more outcomes can deviate.
2. Every joint law is multilinear in the w_k: a sum over persistence patterns of products of frozen-block laws, in which the hidden variable is shared within a block and independent between blocks. The frozen limit ($w_k = 1$) bounds each deviation, and $w_k = 0$ recovers quantum mechanics exactly.
3. Re-preparation acts on the qubit. Whether it also redraws the hidden variable is a separate model choice; the predictions below assume it does not, and Section 7 returns to the alternative.
4. In completion Q the hidden vector also turns with every pulse, including the pulses of a re-preparation, so its predictions depend on the net rotation V that the pulses between two readouts apply. A reset that applies no pulse, such as optical pumping if the model lets it leave $\boldsymbol{\lambda}$ alone, is transparent to $\boldsymbol{\lambda}$. Predictions marked *transparent* assume that the preparation pulse U follows such a reset and that U^{-1} is applied before it, so that $V = UU^{-1}$ is the identity.

Diffusive dynamics (Brownian motion of $\boldsymbol{\lambda}$ on the sphere, or reflecting diffusion of u) share both limits and differ at intermediate persistence; they are not evaluated here.

5 Predictions

Four sequences are considered. *Repeated preparation*: prepare $\boldsymbol{s}$, measure $\boldsymbol{n}$, re-prepare $\boldsymbol{s}$, measure $\boldsymbol{n}$. *Pair*: measure Z, then an axis at angle θ. *Triad*: Z, N_θ, Z with outcomes z_0, r, z_2 (white paper Section 3). *Causal break*: the triad with re-preparation after the first readout (white paper Section 4.4), summarized by the carry-over $D = \frac{1}{2}[\mathbb{E}(z_2|z_0{=}{+}1) - \mathbb{E}(z_2|z_0{=}{-}1)]$. A balanced start, $\boldsymbol{s} = +\hat{\boldsymbol{x}}$, makes the first Z outcome equiprobable.

5.1 Repeated preparation

For a balanced state ($\boldsymbol{s} \perp \boldsymbol{n}$) both completions repeat the first outcome whenever the hidden variable persists, completion Q only with transparent re-preparation:

$$\mathbb{E}(x_1x_2) = w \quad \text{(Q, transparent)}, \qquad \mathbb{E}(x_1x_2) = \sigma_1\sigma_2\, w \quad \text{(D)}, \qquad \mathbb{E}(x_1x_2) = 0 \quad \text{(quantum mechanics)}. \tag{4}$$

The signal is first order in w and needs neither a second basis nor a third readout. It is also the carry-over D of a causal break without the middle readout. Completion D keeps it for any re-preparation, because pulses do not act on the detector variable. For completion Q with balanced readouts and net pulse rotation V, the second readout tests $\boldsymbol{\lambda}$ along $\boldsymbol{n}' = V^{\mathsf{T}}\hat{\boldsymbol{z}}$, so

$$\mathbb{E}(x_1x_2) = w\left(1 - \frac{2\chi}{\pi}\right), \qquad \cos\chi = \hat{\boldsymbol{z}} \cdot V^{\mathsf{T}}\hat{\boldsymbol{z}}. \tag{5}$$

Pulses alone carry the post-readout eigenstate $\pm\hat{\boldsymbol{z}}$ to a state with Z component $\pm\cos\chi$, which is balanced only at $\chi = \pi/2$, where Eq. (5) vanishes. Balanced repeated preparation made of pulses therefore carries no completion-Q signal. The same holds for a reset made of a readout and a conditional π pulse, with or without an inverse pulse before it (checked numerically). Without a reset, re-preparation by pulses is a readout along a turned axis, and completion Q shows the pair deviation of Section 5.3.

5.2 The orthogonal triad

At $\theta = \pi/2$ with a balanced start, both completions give $C_{ZZ} = \mathbb{E}(z_0 z_2) = w_1 w_2$, and the causal break with re-preparation to $+\hat{\boldsymbol{x}}$ gives $D = w_1 w_2$ (completion Q with transparent re-preparation). The triad needs no reset: the pulses R and R^{-1} that set the middle axis return $\boldsymbol{\lambda}$ to its starting orientation, so the triad is the reset-free test of completion Q. The neighboring correlations separate the two: completion D gives $\mathbb{E}(z_0 r) = \sigma w_1$ and $\mathbb{E}(r z_2) = \sigma w_2$, while completion Q gives zero for both, as quantum mechanics does. For a general start with $c_0 = \boldsymbol{s} \cdot \hat{\boldsymbol{z}}$, the frozen endpoint correlation is $1 - |c_0|$ in both completions, whereas any memoryless instrument with a projective first readout gives a value independent of the preparation. With Eq. (3), the triad endpoint is $s_m^2 e^{-\Delta/\tau}$ for $\Delta = \Delta_1 + \Delta_2$, against $s_m e^{-\Delta/\tau}$ for repeated preparation over the same delay; the ratio measures s_m for completion D, and for completion Q with transparent re-preparation.

5.3 Other angles

Table 1 lists the frozen-limit correlations. In completion Q the middle outcome copies the first for $\theta \le \pi/4$ (Appendix A), so the two-readout correlation deviates from $\cos\theta$ by $(1 - \cos\theta)\, w$ at first order. The frozen pair correlation has kinks at $\theta = \pi/4$ and, by symmetry, $3\pi/4$, so the deviation $q(\theta) = \mathbb{E}_{\mathrm{Q}}(z_0 r) - \cos\theta$ is not smooth in the angle (Section 5.5). In completion D with all $\sigma = +1$, the shared u locks every outcome to the label of the first, so all three correlations equal 1 at every angle. The white paper's reference persistence mixture (its Appendix A.3) keeps $\mathbb{E}(z_0 r) = \cos\theta$ and reaches $C_{ZZ} = 1$ at every angle in the same limit. It coincides with completion Q only at $\theta = \pi/2$, so its $\sin^2\theta$ template is the orthogonal slice of this family, not its angle dependence.

	Quantum mechanics		Completion Q			Completion D ($\sigma = +1$)		
θ	$\mathbb{E}(z_0 r)$	C_{ZZ}	$\mathbb{E}(z_0 r)$	$\mathbb{E}(r z_2)$	C_{ZZ}	$\mathbb{E}(z_0 r)$	$\mathbb{E}(r z_2)$	C_{ZZ}
0	1	1	1	1	1	1	1	1
$\pi/8$	0.924	0.854	1	1	1	1	1	1
$\pi/4$	0.707	0.500	1	1	1	1	1	1
$3\pi/8$	0.383	0.146	0.613	0.865	0.747	1	1	1
$\pi/2$	0	0	0	0	1	1	1	1

Table 1: Frozen-limit ($w_1 = w_2 = 1$) correlations of the triad $Z \to N_\theta \to Z$ with a balanced start ($\boldsymbol{s} = +\hat{\boldsymbol{x}}$, first outcome equiprobable), for the two completions of Section 3; completion D uses the same bright mapping ($\sigma = +1$) for all three readouts. Values are exact for D and from quadrature accurate to 10^{-6} for Q, computed by `analysis/sequential_dh_model.py`. At persistence $w_k < 1$ each entry moves toward the quantum value multilinearly (Section 4).

5.4 Randomized readout labels

Suppose each readout's bright mapping σ is drawn independently and uniformly, implemented as an optional π pulse before detection and undone in software. Several properties follow (checked numerically for the cases stated):

1. Every completion-Q prediction is unchanged in frame-tracked labels. Equation (2) does not involve σ, but the optional π pulse turns $\boldsymbol{\lambda}$ with the qubit; if it is not undone, it multiplies every later completion-Q correlation with that readout by σ, and the analysis multiplies it back.
2. Completion D vanishes exactly from balanced repeated preparation and from all three correlations of the balanced orthogonal triad, and the neighboring correlations of the balanced triad return to $\cos\theta$ at every angle. At intermediate angles the frozen endpoint becomes $\cos\theta$ instead of $\cos^2\theta$, a residual $\cos\theta(1-\cos\theta)$ of at most $1/4$.
3. Conventional memory carried by the detector, the motion, or anything else that registers only the physical bright or dark result is removed from balanced repeated preparation in the same way (Proposition 1).

Proposition 1. *Let the second readout of a repeated-preparation record be any two-outcome measurement whose bright effect $E_M = a(M)\,I + \boldsymbol{b}(M)\cdot\boldsymbol{\sigma}$ may depend on the memory M carried by the apparatus, and let the re-prepared qubit have logical Bloch vector $\boldsymbol{r}(M)$. If the relabeling bit σ_2 is independent of M and the logical outcome is $x_2 = \sigma_2 \times$ (physical outcome), then*

$$\mathbb{E}(x_2 \mid M) = 2\big[b_y(M)\,r_y(M) + b_z(M)\,r_z(M)\big]. \tag{6}$$

Proof. A π pulse about x maps $(r_x, r_y, r_z) \mapsto (r_x, -r_y, -r_z)$, and the physical outcome has mean $2a - 1 + 2\boldsymbol{b}\cdot\boldsymbol{r}_{\text{phys}}$. Multiplying by σ_2 and averaging over $\sigma_2 = \pm 1$ leaves only the terms even in σ_2. □

Two consequences follow. If re-preparation yields $+\hat{\boldsymbol{x}}$ whatever M holds, memory of any form in the apparatus produces no logical-label correlation. A residual requires the re-prepared state itself to depend on the first readout, and with σ_1 also random it survives only if that dependence follows the logical outcome $\sigma_1 \times$ (physical outcome). The apparatus registers the physical result and the pulse separately; only the qubit (a nonprojective first readout followed by an incomplete reset) or a controller that computes logical labels can carry their product. Completion Q evades Eq. (6) because its outcome depends on $\boldsymbol{\lambda}$ beyond the quantum state. Because $\boldsymbol{\lambda}$ turns with the qubit, completion-Q persistence follows the qubit's physical frame, as the residual of an incomplete reset does. Relabeling therefore separates completion Q from memory carried by the detector, but not from memory carried by the qubit; a verified reset bounds the latter, and the middle readout of the orthogonal triad removes it.

5.5 A shape test

The correlations above have a size but no shape that conventional memory could not match. A scan of the second preparation supplies one. In repeated preparation with a balanced first state and the second state at polar angle ϕ, both completions give

$$\mathrm{Cov}(x_1, x_2) = w\,(1 - |\cos\phi|), \qquad \mathbb{E}(x_2) = \cos\phi, \tag{7}$$

with fixed labels, completion Q with transparent re-preparation. The shared threshold couples the two outcomes as tightly as their marginals allow, $\mathbb{P}(x_1{=}x_2{=}{+}1) = \min(p_1, p_2)$ in the frozen limit, and the minimum has a cusp where the two probabilities cross, here at $\phi = \pi/2$. The marginal stays at its Born value. When re-preparation is made of pulses, as in a reset-free chain in which two $\pi/2$ pulses with relative frame phase ϕ turn the readout axis by θ with $\cos\theta = -\cos\phi$, completion D keeps the cusp on top of the quantum correlation $\cos\theta$, and completion Q shows instead the deviation $w\,q(\theta)$ of Section 5.3, with its kinks.

Proposition 2. *Let ϕ enter a record only through one frame phase $e^{-i\phi\hat{n}}$, with $\hat{n} = \sum_j j\,|j\rangle\langle j|$ on the d levels the drive reaches, and let every other operation (preparations, instruments, resets, and memory in the apparatus, the motion or the controller) be independent of ϕ. Then every outcome probability is a trigonometric polynomial in ϕ of degree at most $d-1$.*

Proof. The probability is linear in the state after the phase, and conjugation by $e^{-i\phi\hat{n}}$ multiplies $|j\rangle\langle k|$, tensored with any operator on other systems, by $e^{-i\phi(j-k)}$ with $|j-k| \le d-1$. □

On platforms with virtual Z rotations, the frame phase shifts the phase of every later pulse instead of acting on the qubit, so a shifted pulse P becomes $e^{-i\phi\hat{n}}Pe^{i\phi\hat{n}}$. If the readout that ends the record is diagonal in the level basis, as fluorescence and dispersive readouts are to good approximation, the outer factor commutes with it and the record's probabilities see one conjugation; for a general readout the bound doubles to $2(d-1)$. Drawing ϕ at random for each record keeps memory of earlier records from carrying it. A qubit with one leakage level has $d = 3$, so the degree is at most 2, or 4 without the diagonal readout. The cusp instead contains every even harmonic, $1 - |\cos\phi| = 1 - \frac{2}{\pi} - \frac{4}{\pi}\sum_{k\geq 1}(-1)^{k+1}\cos(2k\phi)/(4k^2-1)$. Several consequences follow:

1. Persistence of a threshold hidden variable produces harmonics that no quantum model with memory produces under these conditions. Above the second harmonic the cusp carries an rms amplitude of $0.068\,w$; above the fourth, $0.032\,w$. The completion-Q deviation q contains only odd harmonics and is dominated by the third; it carries $0.137\,w$ above the second harmonic and $0.017\,w$ above the fourth.
2. A template fit after removing harmonics up to second order identifies w at five standard deviations when $w \geq 5/(0.068\sqrt{N})$: 0.073, 0.023 and 0.007 for $N = 10^6$, 10^7 and 10^8 records. Removing harmonics up to fourth order raises these to 0.15, 0.049 and 0.015. For q the thresholds are 0.037, 0.012 and 0.004 above the second harmonic, and 0.30, 0.094 and 0.030 above the fourth. The fixed-angle correlation detects $w \approx 0.003$ with 10^7 records, so the shape test is a second step, after a detection.
3. Random relabeling places the completion-Q signal in frame-tracked labels and the completion-D cusp in physical labels. In physical labels the cusp separates completion D from conventional detector memory, which Proposition 2 confines to low-order harmonics.
4. Imperfections in generating the phase, such as mixer imbalance or converter nonlinearity, can add small higher harmonics. The single-readout marginal, which both completions keep exactly at $\cos\phi$, measures them in the same data.

Long persistence also permits a direct readout of the hidden variable. If it survives many records, stepping the preparation probability up and down within one persistence window gives outcomes that switch once and never switch back, at the threshold u for completion D or at a projection of $\boldsymbol{\lambda}$ for completion Q; quantum mechanics gives independent outcomes. Window-by-window threshold estimates should then be uniformly distributed, and their autocorrelation measures τ directly. This requires w close to 1 across a window.

5.6 What the two anchorings imply

1. Qubit-anchored persistence follows the qubit's frame, as memory carried by the qubit does. In the orthogonal triad the middle readout projects the qubit, so quantum mechanics with a memoryless readout makes the endpoints independent given the middle outcome, while completion Q correlates them at $w_1 w_2$ in frame-tracked labels. No conventional counterpart to that endpoint correlation has been identified here. A positive result would be a candidate anomaly requiring replication on a second platform.
2. Detector-anchored persistence shares its fixed-angle signature with conventional detector and motional memory, so at fixed angles only a null result is informative: it bounds w for that detector at that delay. The cusp of Section 5.5 separates the two, at a higher threshold in w.
3. Same-basis readouts of a qubit already in an eigenstate ($\theta = 0$ without re-preparation) carry no signature in either completion, since quantum mechanics and both completions give $r = z_0$.

6 Existing data that may constrain the family

1. *Detector-anchored persistence* is probed by any repeated measurement of identically prepared balanced states with the same detector, including discrete-outcome quantum random-number generators. Raw-bit serial correlations reported for such devices bound w at their repetition intervals for their detectors. Three examples show what exists:

(a) A which-path generator with one uncooled silicon avalanche photodiode at about 10^5 bits per second showed, in 10^9 raw bits, a lag-1 deviation of about 2×10^{-4} toward equal neighbors, attributed to detector recovery [11].
(b) A beam-splitter generator with two cooled single-photon avalanche diodes gave raw lag-1 coefficients from +0.030 at 1 kHz per detector to −0.134 at 10 MHz; afterpulsing supplies the positive part, at about the afterpulse probability [12].
(c) Generators on cloud superconducting qubits report pass rates of serial tests rather than coefficients [13].

Afterpulsing produces exactly the positive correlation that detector-anchored persistence predicts. These records therefore bound w only to the level of the detector effect, from about 10^{-4} to a few 10^{-2} depending on detector and rate, and only if the two do not cancel. The bound is also detector-specific, because the timescale heuristic ties τ to detector volume and temperature (white paper Section 3.4).

2. *Qubit-anchored persistence* requires the same system across measurements. Sequential single-ion measurements [14, 15] that agree with quantum two-time correlations at the percent level would bound w at the percent level at their delays, through the first-order deviation $(1 - \cos\theta)\, w$ for $\theta \le \pi/4$. A bound requires the raw records and their calibration.
3. *Two-party persistence* would appear as serial correlations at fixed settings in the time-tagged records that loophole-free Bell tests have published. Predicting their size needs the Bell-scenario extension of Section 8.

7 Consequences for the experiment

Within this family, the white paper's program changes as follows:

1. For completion D the first test is balanced repeated preparation with two readouts per record, which needs no particular re-preparation. Its statistic has standard error $1/\sqrt{N}$, so 2.5×10^5 records bound $w < 0.01$ at five standard deviations. For completion Q the first test is the orthogonal triad, or a chain of alternating orthogonal readouts, whose frame-tracked endpoint statistic has the same standard error and bounds $w_1 w_2$. Balanced repeated preparation tests completion Q only with transparent re-preparation.
2. Randomizing each readout's bright mapping separates the two anchorings and removes conventional detector memory from the logical-label statistic (Section 5.4). It costs one optional π pulse per readout. The post-readout state should go to reset without first undoing that pulse, so that nothing in the qubit follows the logical label; the analysis then tracks the frame that each pulse leaves on $\boldsymbol{\lambda}$. On platforms with virtual Z rotations the relabeling can instead be a frame phase on the preparation pulses, which leaves the pulses themselves unchanged.
3. The orthogonal triad measures $s_m^2 e^{-\Delta/\tau}$. Compared with repeated preparation at matched total delay it isolates the per-measurement survival s_m, and as a reset-free sequence it covers the cases in which re-preparation redraws or turns the hidden variable.
4. Angle-resolved analysis should fit the completions directly, because the $\sin^2\theta$ template is only their orthogonal slice (Table 1).
5. If the first test shows a correlation, a scan of the second preparation angle, set by a frame phase drawn at random for each record, tests for the cusp of Eq. (7) or the kinks of completion Q. With 10^7 to 10^8 records it identifies persistence down to $w \approx 0.02$ to 0.007 for the cusp and 0.012 to 0.004 for the kinks.
6. A design for IBM's cloud superconducting processors is in the package's `notes/CLOUD_RUN_DESIGN.md`. It uses reset-free chains whose settings are all frame phases, and a pilot and a main job within the free allowance. Nothing has been run.

8 Open questions

1. *Bell scenario.* A two-party version in which the hidden variable's distribution depends on both settings would make the persistence superdeterministic in the sense of the parent model and would predict serial correlations

in Bell-test records.

2. *Dynamics.* Renewal and diffusion differ at intermediate persistence, and neither identifies the physical degree of freedom that sets τ and s_m.
3. *Acceptance.* Whether the authors of the parent model would accept either completion is open; their answer would decide whether a bound on this family bears on their proposal.

A Frozen-limit derivations

Write $a = \boldsymbol{\lambda} \cdot \hat{\boldsymbol{z}}$, $b = \boldsymbol{\lambda} \cdot \boldsymbol{n}$, $c = \cos\theta$ and $s = \sin\theta$. For isotropic $\boldsymbol{\lambda}$, a is uniform on $[-1, 1]$, and given a the component λ_x follows the arcsine law on $[-\rho, \rho]$ with $\rho = \sqrt{1 - a^2}$.

Orthogonal triad, general start. At $\theta = \pi/2$, completion Q gives $z_0 = \mathrm{sgn}(a + c_0)$, $r = \mathrm{sgn}(b)$ and $z_2 = \mathrm{sgn}(a)$. The endpoints differ only for a between 0 and $-c_0$, so $C_{ZZ} = 1 - |c_0|$; reflecting $\boldsymbol{\lambda}$ through the plane normal to $\boldsymbol{n}$ reverses b at fixed a, so $\mathbb{E}(z_0 r) = \mathbb{E}(r z_2) = 0$. In completion D with bright mappings $\sigma_0, \sigma, \sigma_2$, both later thresholds equal $1/2$, so $r = \sigma\ \mathrm{sgn}(\frac{1}{2} - u)$ and $z_2 = \sigma_2\,\mathrm{sgn}(\frac{1}{2} - u)$, while $z_0 = \sigma_0\,\mathrm{sgn}(p_b - u)$; this gives $C_{ZZ} = \sigma_0\sigma_2(1 - |c_0|)$, $\mathbb{E}(z_0 r) = \sigma_0\sigma(1 - |c_0|)$ and $\mathbb{E}(r z_2) = \sigma\sigma_2$.

Pair, completion Q. With a balanced start, $z_0 = \mathrm{sgn}(a)$ and $r = \mathrm{sgn}(b + z_0 c)$. For $z_0 = +1$, $b + c = c(1 + a) + s\lambda_x \geq c - s\rho \geq 0$ whenever $c \geq s$, so $r = z_0$ for every $\theta \leq \pi/4$. In general $\mathbb{E}(z_0 r) = 1 - 4\mathbb{P}(a > 0,\ b < -c)$ with

$$\mathbb{P}(a > 0,\ b < -c) = \frac{1}{2\pi}\int_0^{a_*} \arccos\left(\frac{c}{s}\sqrt{\frac{1+a}{1-a}}\right) \mathrm{d}a, \qquad a_* = \max\bigl(0,\ s^2 - c^2\bigr), \tag{8}$$

and the triad endpoint is $C_{ZZ} = 1 - 4\mathbb{P}(0 < a < c,\ b < -c)$, the same integral with upper limit $\min(c, a_*)$.

B Numerical checks

`analysis/sequential_dh_model.py` evaluates frozen blocks exactly for completion D (interval arithmetic in u) and by quadrature in a with the exact arcsine law in λ_x for completion Q, with panel edges at every square-root kink. It checks, in 718 cases: Born marginals; recovery of the ideal triplet law and of zero carry-over at $w = 0$; the closed forms above for general starts and both bright mappings; the renewal results of Section 5; the relabeling results of Section 5.4; the pair and triad integrals; the cusp of Eq. (7) in both label frames with fixed and random labels; Proposition 2 on 60 random quantum models with a two-level memory ($d = 2$ and $d = 3$; the phase as one rotation, as shifted pulses before a level-diagonal readout, and as shifted pulses before a general readout), where harmonics above the stated bound vanish to 10^{-12} and the top allowed harmonic is present; the cusp's harmonic content against its Fourier series; Eq. (5) for completion Q and the axis independence of completion D; frame tracking of completion Q under relabeling pulses that are not undone; the odd-only harmonic content of the deviation q on two grids; and Monte Carlo agreement within five standard errors at 4×10^5 records per case. It validates formulas only.

References

[1] Sandro Donadi and Sabine Hossenfelder. Toy model for local and deterministic wave-function collapse. *Physical Review A*, 106:022212, 2022. doi: 10.1103/PhysRevA.106.022212. URL https://arxiv.org/abs/2010.01327.

[2] Sabine Hossenfelder. Testing super-deterministic hidden variables theories. *Foundations of Physics*, 41:1521–1531, 2011. doi: 10.1007/s10701-011-9565-0. URL https://arxiv.org/abs/1105.4326.

[3] Sabine Hossenfelder. Testing Superdeterministic Conspiracy. *Journal of Physics: Conference Series*, 504:012018, 2014. doi: 10.1088/1742-6596/504/1/012018. URL https://arxiv.org/abs/1401.0286.

[4] Jonte R. Hance and Sabine Hossenfelder. Comment on "Experimentally adjudicating between different causal accounts of Bell-inequality violations via statistical model selection". *Physical Review A*, 109:026201, 2024. doi: 10.1103/PhysRevA.109.026201. URL https://arxiv.org/abs/2206.10619.

[5] Sabine Hossenfelder. How Gravity Can Explain the Collapse of the Wavefunction, 2025. URL https://arxiv.org/abs/2510.11037. arXiv preprint, version 1.

[6] N. David Mermin. Hidden variables and the two theorems of John Bell. *Reviews of Modern Physics*, 65:803–815, 1993. doi: 10.1103/RevModPhys.65.803. URL https://arxiv.org/abs/1802.10119.

[7] Nicholas Harrigan and Robert W. Spekkens. Einstein, incompleteness, and the epistemic view of quantum states. *Foundations of Physics*, 40:125–157, 2010. doi: 10.1007/s10701-009-9347-0. URL https://arxiv.org/abs/0706.2661.

[8] John S. Bell. On the problem of hidden variables in quantum mechanics. *Reviews of Modern Physics*, 38:447–452, 1966. doi: 10.1103/RevModPhys.38.447.

[9] Simon Kochen and E. P. Specker. The problem of hidden variables in quantum mechanics. *Journal of Mathematics and Mechanics*, 17:59–87, 1967. doi: 10.1512/iumj.1968.17.17004.

[10] T. Erber, P. Hammerling, G. Hockney, M. Porrati, and S. Putterman. Resonance fluorescence and quantum jumps in single atoms: Testing the randomness of quantum mechanics. *Annals of Physics*, 190:254–309, 1989. doi: 10.1016/0003-4916(89)90016-X. URL https://www.sciencedirect.com/science/article/pii/000349168990016X.

[11] André Stefanov, Nicolas Gisin, Olivier Guinnard, Laurent Guinnard, and Hugo Zbinden. Optical quantum random number generator. *Journal of Modern Optics*, 47:595–598, 2000. doi: 10.1080/09500340008233380. URL https://arxiv.org/abs/quant-ph/9907006.

[12] Mario Stipčević and Daniel J. Gauthier. Precise Monte Carlo simulation of single-photon detectors. In *Proceedings of SPIE*, volume 8727, page 87270K, 2013. doi: 10.1117/12.2021577. URL https://arxiv.org/abs/1411.3663.

[13] Yutaka Shikano, Kentaro Tamura, and Rudy Raymond. Detecting temporal correlation via quantum random number generation. *Electronic Proceedings in Theoretical Computer Science*, 315:18–25, 2020. doi: 10.4204/EPTCS.315.2. URL https://arxiv.org/abs/2004.01330.

[14] G. Kirchmair, F. Zähringer, R. Gerritsma, M. Kleinmann, O. Gühne, A. Cabello, R. Blatt, and C. F. Roos. State-independent experimental test of quantum contextuality. *Nature*, 460:494–497, 2009. doi: 10.1038/nature08172. URL https://arxiv.org/abs/0904.1655.

[15] F. M. Leupold, M. Malinowski, C. Zhang, V. Negnevitsky, A. Cabello, J. Alonso, and J. P. Home. Sustained State-Independent Quantum Contextual Correlations from a Single Ion. *Physical Review Letters*, 120:180401, 2018. doi: 10.1103/PhysRevLett.120.180401. URL https://arxiv.org/abs/1706.07370.